

Overcoming Shortcut Problem in VLM for Robust Out-of-Distribution Detection

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Abstract

Vision-language models (VLMs), such as CLIP, have shown remarkable capabilities in downstream tasks. However, the coupling of semantic information between the foreground and the background in images leads to significant shortcut issues that adversely affect out-of-distribution (OOD) detection abilities. When confronted with a background OOD sample, VLMs are prone to misidentifying it as in-distribution (ID) data. In this paper, we analyze the OOD problem from the perspective of shortcuts in VLMs and propose OSPCoOp which includes background decoupling and mask-guided region regularization. We first decouple images into ID-relevant and ID-irrelevant regions and utilize the latter to generate a large number of augmented OOD background samples as pseudo-OOD supervision. We then use the masks from background decoupling to adjust the model's attention, minimizing its focus on ID-irrelevant regions. To assess the model's robustness against background interference, we introduce a new OOD evaluation dataset, ImageNet-Bg, which solely consists of background images with all ID-relevant regions removed. Our method demonstrates exceptional performance in few-shot scenarios, achieving strong results even in one-shot setting, and outperforms existing methods. The code and proposed ImageNet-Bg are available at https://github.com/HAIV-Lab/OSPCoOp_Imagenet-bg.

1. Introduction

In open-world scenarios [3, 22], the deployed machine learning models may encounter out-of-distribution (OOD) samples [17] not present during training. Incorrect predictions on OOD samples can lead to serious consequences, especially in critical fields like autonomous driving [12]. Thus, the ability to detect OOD samples is crucial.

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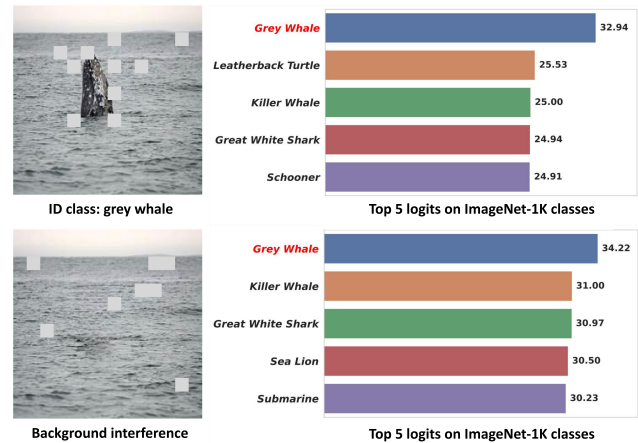


Figure 1. The significant shortcut issue with CLIP. The gray patches in the image possess the highest similarity to the ground truth. CLIP classifies the image solely based on the background region. Ridiculously, when the ID-relevant regions are removed, CLIP produces higher logits on the background interference.

Traditional OOD detection methods primarily focus on single-modal models [9, 17, 18, 30, 31, 46, 48, 56], which are trained from scratch in a fully supervised manner, requiring substantial amounts of data and resources. Recently, vision-language models (VLMs) [26, 40, 53] like CLIP [40] have demonstrated remarkable zero-shot performance and can be applied to various downstream tasks with minimal parameter tuning [20, 60]. However, despite their extensive semantic understanding, VLMs still encounter challenges with OOD data, as it is impractical to enumerate every label these models may encounter [33, 57]. Consequently, OOD detection based on VLMs is attracting increasing attention.

Previous studies on CLIP-based OOD detection can be categorized into two main types: training-free and training-required methods [37]. Training-free methods leverage CLIP's incredible zero-shot capabilities for OOD detection [21, 33]. However, their performance is restricted by the fixed feature space of CLIP and the manually designed prompts. In contrast, training-required methods [27, 38] in-

volves fine-tuning specific parameters of CLIP to improve the discrimination between ID and OOD samples. However, we discover that CLIP is confronted with a serious shortcut problem, which can have catastrophic effects on its ability to detect OOD samples. Regrettably, existing methods to a large extent neglect this critical problem.

In deep neural networks (DNNs) [16, 25], models often learn shortcuts through two main aspects [15]. First, shortcut opportunities arise when familiar backgrounds become as significant, or even more so, than the objects themselves. Second, models may rely on specific or irrelevant features for recognition. We focus primarily on the first aspect, as shortcut opportunities are particularly prevalent in CLIP models that are pre-trained on image-text pairs. The coupling of foreground and background descriptions in the pre-training text can create misleading correlations between background text and foreground regions, leading to incorrect classifications of background interference as ID categories. Fig. 1 illustrates the detrimental impact of the shortcut problem on background interference.

To address this issue, we propose OSPCoOp with background decoupling and mask-guided region regularization. Motivated by [28, 41], which uses the inpainting models [45, 52] to edit background regions and generate diverse background images with the same foreground for data augmentation, we apply the segmentation [24, 32, 42] models to remove the foreground regions of images and inpainting them with the background information to generate pure background images as pseudo-OOD supervision. Besides, we also use the masks generated from the segmentation model to constrain the regions of the images and suppress the model’s local logits in those areas. Unlike LoCoOp [35], which directly uses the similarity of CLIP features to delineate ID-irrelevant regions and results in a significant number of background regions being incorrectly classified as ID-relevant due to the shortcut problem in CLIP, we use mask instead to define these regions directly. Additionally, to assess the model’s robustness against background interference, we propose a new OOD evaluation dataset named ImageNet-Bg. All images in this dataset are derived from the ImageNet validation set [7] and solely consist of backgrounds, with all ID-relevant regions removed.

Our main contributions can be summarized as follows:

- We find that CLIP suffers from serious shortcut problem, resulting in the misclassification of background images as ID samples. To address this, we propose background decoupling to obtain pseudo-OOD supervision and use decoupled background replacement to enhance ID images.
- To reduce excessive attention to background regions due to the shortcut problem in CLIP, we propose mask-guided region regularization to constrain ID-irrelevant areas and reduce the model’s response to these regions.
- We propose the ImageNet-Bg OOD test set, which con-

sists of ImageNet validation images with ID-relevant regions removed, facilitating a comprehensive evaluation of the model’s robustness against background interference.

2. Related Work

2.1. Training-free OOD Detection with VLMs

Training-free OOD detection methods leverage the powerful representational capabilities of pre-trained VLMs, e.g. CLIP. These works often focus on designing OOD scores that better differentiate between ID and OOD samples.

Fort et al. [13], ZOC [10] and MCM [34] are early works that employ VLMs for OOD detection. GL-MCM [39] enhances MCM with local visual information. CLIPN [49] calculates the OOD score using the pre-trained negative text encoders. DPM [57] finds that solely using textual matching overlooks significant ID information and proposes that matching visual patterns enhances OOD performance. SeTAR [29] uses selective low-rank approximation of weight matrices in CLIP to enhance OOD Detection performance.

Recently some works have explored outlier exposure for VLMs. NegLabel [21] designs an OOD score in combination with negative labels from WordNet [11]. AdaNeg [54] generates adaptive negative proxies during testing. CLIP-Scope [14] improves the NegLabel Score via Bayesian inference. OLE [8] utilizes outlier labels to execute outlier prototype learning. EOE [4] exploits the expert knowledge within LLMs [1] to envision potential outlier labels.

2.2. Few-Shot OOD Detection with VLMs

Considering that fully supervised tuning of VLMs requires substantial computational resources, Miyai et al. [35] firstly present the few-shot OOD detection setting. They propose LoCoOp, a simple yet effective method that guides the model’s attention to ID-relevant regions. However, it is often seriously interfered by the background, which leads to inaccurate location of the ID-relevant regions. SCT [51] further refines LoCoOp by introducing modulating factors for the loss items but has only a slight improvement.

Recently, some works have focused on constructing OOD samples and OOD prompts. NegPrompt [27] learns negative prompts for VLMs to represent negative semantics relative to ID. Bai et al. [2] introduce ID-like prompts and construct outliers highly correlated with ID data. LAPT [55] fully automates the collection of labels and images for prompt tuning. CLIP-OS [43] uses CLIP-surgery-discrepancy Masking to synthesize reliable OOD data and proposes an ‘unknown’ prompt to regulate the ID/OOD boundary. This method shares similarities with our idea. However, the ID and OOD samples obtained by this method are neither accurate nor natural. Moreover, lumping all OOD data into a single ‘unknown’ category renders different OOD classes indistinguishable from one another.

2.3. Prompt Learning for VLMs

Prompt learning initially aimed to automate the creation of templates for extracting knowledge from LLMs and VLMs. It utilizes supervised learning to streamline prompt development, reducing the need for manual input. CoOp [60] employs learnable vectors for context words while keeping the VLM parameters fixed. CoCoOp [59] enhances this method by introducing a lightweight neural network to generate input-conditional tokens for each image. Dual-CoOp [44] develops a method to learn pairs of positive and negative prompts. There are also works such as VPT [20] and MAPLE [23] that insert prompts into visual encoders. However, due to the shortcut problem in VLMs, the learned prompts are likely to pay excessive attention to the background, which can be detrimental to downstream tasks.

3. Methodology

3.1. Preliminaries

Problem statement. For OOD detection with VLMs, an OOD detector is a binary classifier for ID and OOD images. Assume an ID dataset D^{id} of (x^{id}, y^{id}) pairs with x^{id} as ID image and $y^{id} \in Y^{id} := 1, \dots, K$ as ID class label. Assume an OOD dataset D^{ood} of (x^{ood}, y^{ood}) pairs with x^{ood} as OOD image and $y^{ood} \in Y^{ood} := K + 1, \dots, K + O$ as OOD class label. There are no class overlaps between ID and OOD datasets. We focus on the few-shot scenario where the model is fine-tuned on D_{train}^{id} with few samples per class and the test set contains D_{test}^{id} and D_{test}^{ood} .

VLM-based classification. VLMs represented by CLIP include an image encoder $\mathcal{I} : x \rightarrow \mathbb{R}^{H \times W \times D}$ and a text encoder $\mathcal{T} : t \rightarrow \mathbb{R}^{1 \times D}$, which are used to obtain the region image feature f^v of image x and text feature $\mathcal{T}(t)$ of text t . Then the region image feature is pooled as global image feature $F^V \in \mathbb{R}^{1 \times D}$. For a downstream classification task with a label set $\mathcal{Y} = \{y_1, y_2, \dots, y_K\}$ containing K classes, we can obtain a collection of text features $\mathcal{T}(t_k)$, where t_k is the prompt for a label y_k . The predicted probability p_k corresponding to y_k can be obtained by:

$$p(y_k | \mathbf{x}) = \frac{\exp(\text{sim}(F^V, \mathcal{T}(t_k)) / \tau)}{\sum_{j=1}^K \exp(\text{sim}(F^V, \mathcal{T}(t_j)) / \tau)}, \quad (1)$$

where sim denotes the cosine similarity and τ is the temperature of softmax.

Prompt learning. To further improve the performance of VLMs on downstream tasks, several prompt learning methods are proposed. The textual prompt is formulated as $t_k = \{\omega_1, \omega_2, \dots, \omega_N, c_k\}$, where c_k denotes the word embedding of the training class name, $\{\omega_n\}_{n=1}^N$ are learnable vectors, and N denotes the length of learnable context words. With prompt t_k as input, the text encoder $\mathcal{T}$ outputs the textual feature as $\mathcal{T}(t_k)$. The final prediction probability

is computed by the matching score as eq 1 and is used to calculate cross-entropy loss for backpropagation.

Local regularized prompt learning for OOD detection. To further utilize prompt learning in few-shot OOD detection, LoCoOp [35] proposes Local Regularized Prompt Learning, extracting ID-irrelevant regions from CLIP local features and performing OOD regularization training with the extracted regions. The loss function for this regularization is $\mathcal{L}_{ood} = -H(p_k)$, where $H(\cdot)$ is the entropy function and p_k denotes the classification prediction probabilities for ID-irrelevant regions.

3.2. Decoupling Background for Augmentation

VLMs represented by CLIP often take shortcuts when classifying, i.e. they may rely more on the ID-irrelevant background rather than the ID-relevant foreground. To address this issue, we decouple the images into foreground and background for data augmentation.

Background decoupling. To suppress model's response to the background interference, we first decouple the images into foreground and background, the latter of which contains no ID information and can be treated as OOD samples. Specifically, for an ID sample, we first segment the ID-relevant mask m^{id} from the ID samples (x^{id}, y^{id}) with segment models denoted as $\text{Seg}(\cdot)$. To maintain the integrity of the mask, we use dilation kernel B on the mask to ensure it completely covers the ID-relevant regions in the image. Then we remove the ID-relevant regions and inpaint the masked regions with background information. Supposing $\text{Inp}(\cdot)$ as the image inpainting models and the inpainted images with the ID-relevant contents removed x^{ood} can be obtained:

$$\begin{aligned} m^{id} &= \text{Seg}(x^{id}, y^{id}), \\ x^{ood} &= \text{Inp}(x^{id} \odot (1 - m^{id}), m^{id} \oplus B). \end{aligned} \quad (2)$$

However, the $\text{Seg}(\cdot)$ may still not completely segment the ID regions, leading to x^{ood} retaining some ID-relevant contents, we utilize the pre-trained CLIP model to filter x^{ood} based on their similarity with the ground truth y_k of x^{id} to provide pseudo-OOD supervision with pure background:

$$x^{ood} = \{x^{ood} | \epsilon < z_k^{id} - z_k^{ood}\}, z_k = \text{sim}(\mathcal{I}(x), \mathcal{T}(t_k)), \quad (3)$$

where ϵ stands for the similarity thresholds.

OOD image augmentation. We note that when considering the local features of ID samples, they are very likely to be difficult to distinguish from the local features of several OOD samples, especially those OOD samples with significant textures. To address this issue, we also construct some OOD samples by repeating the local texture of the ID-relevant region. It can also be denoted as x^{ood} . Some examples of constructed OOD samples are shown in Fig. 2. All the constructed OOD samples form the set D_{train}^{ood} .

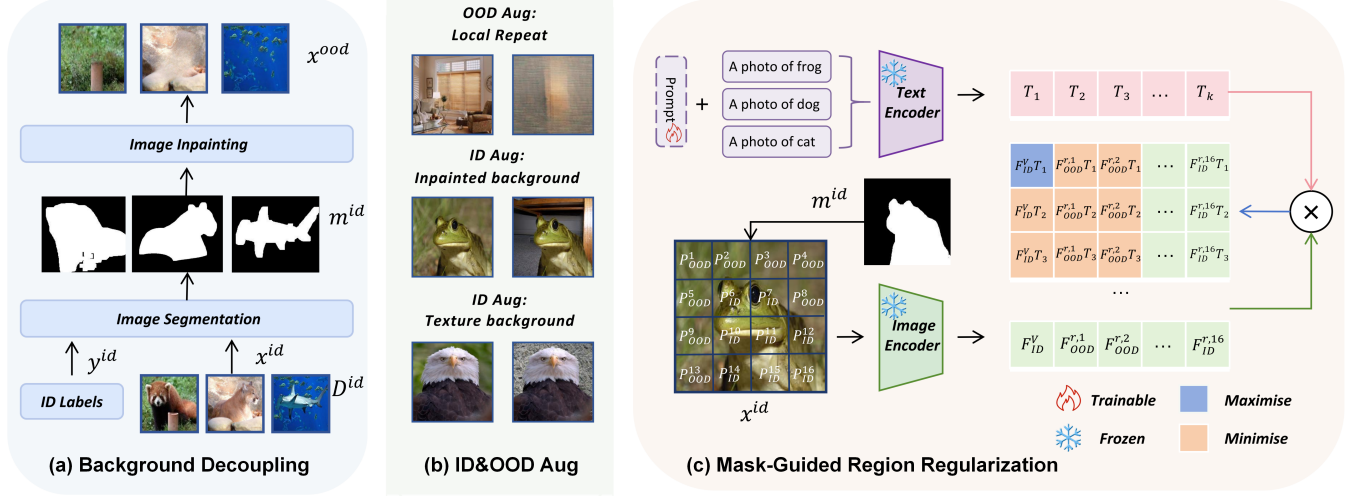


Figure 2. Pipeline of OSPCoOp. We first decouple the images into foreground and background to provide pseudo-OOO supervision. Subsequently, we generate ID and pseudo-OOO augmented samples and then use the masks obtained during the background decoupling process to provide regional supervision for ID and OOD samples, guiding the model to ignore the ID-irrelevant regions.

ID image augmentation. To compel the VLMs to pay more attention to the ID-relevant region, we also perform augmentations on the ID samples by replacing the irrelevant background regions with inpainted OOD samples and augmented texture OOD samples. Some examples of augmented ID samples are shown in Fig. 2. All the augmented ID samples and raw ID samples form the set D_{train}^{id} .

3.3. Mask-Guided Region Regularization

For each ID sample x^{id} , we have obtained its corresponding inpainted background image x^{ood} and mask for ID-relevant region m^{id} mentioned in Sec. 3.2. Let $\mathcal{P}$ be the set of all patches in the image. For each patch $P \in \mathcal{P}$ of the image x^{id} , we count the occurrences of m^{id} within each patch to classify the patches as ID-relevant patches P^{id} or ID-irrelevant patches P^{ood} :

$$P \in \begin{cases} P^{id} & \text{if } \sum_{j=0}^n m_j^{id} > \theta \\ P^{ood} & \text{if } \sum_{j=0}^n m_j^{id} \leq \theta, \end{cases} \quad (4)$$

where n is the number of pixels contained in the patch, and θ is the threshold, typically set to $\frac{n}{2}$. For all patches belonging to the inpainted background images and augmented ood images x^{ood} , they are classified as ID-irrelevant patches P^{ood} , as all of their ID-relevant content has been removed. Based on the ID relevance division of patches, we can obtain the corresponding ID relevance division of the region image features. For the image x^{id} containing N ID-relevant patches, we can obtain its ID-relevant global visual features $F_{ID}^V \in \mathbb{R}^{1 \times D}$, ID-relevant region image features $F_{ID}^R \in \mathbb{R}^{N \times D}$, and ID-irrelevant region image features $F_{OOD}^R \in \mathbb{R}^{(H \times W - N) \times D}$. For the image x^{ood} that

does not contain any ID-relevant patches, we can obtain its ID-irrelevant global visual features $F_{OOD}^V \in \mathbb{R}^{1 \times D}$ and ID-irrelevant regional visual features $F_{OOD}^R \in \mathbb{R}^{(H \times W) \times D}$. During the training process, we optimize the similarity between each F_{ID}^V and their corresponding ID text features $\mathcal{T}(t_k)$, while suppressing the similarity between each F_{OOD}^R , F_{OOD}^V with all ID category text features. The entire optimization process is completed using the following loss. **ID-global relevance loss.** To enhance the model’s ability to learn features relevant to specific ID classes and improve classification accuracy, we employ the standard cross-entropy loss to optimize the divergence between the predicted label probabilities and the corresponding ground truth labels for the ID sample x^{id} . The ID-global relevance loss $\mathcal{L}_{in}$ is defined as:

$$\mathcal{L}_{id} = -\frac{1}{|D_{train}^{id}|} \sum_{x^{id} \in D_{train}^{id}} \log \frac{\exp(z_k/\tau)}{\sum_{j=1}^K \exp(z_j/\tau)}. \quad (5)$$

OOD-region irrelevance loss. To address the shortcut problem in the CLIP model and reduce its excessive focus on background information, we calculate the negative value of the entropy of all ID-irrelevant regions as the OOD-region irrelevance loss. When the predicted probability distribution of ID-irrelevant regions across all ID categories is uniform (with values close to each other), the model exhibits uncertainty in its predictions for ID-irrelevant regions, resulting in a low loss value. The OOD-region irrelevance loss $\mathcal{L}_{ood}^r$ is defined as follows:

$$\mathcal{L}_{ood}^r = -\frac{1}{|D_{train}^{all}|} \sum_{x^{id} \in D_{train}^{all}} \frac{1}{N_i} \sum_{n=1}^{N_i} H[p(y | P_n^{ood})], \quad (6)$$



Figure 3. Some images from ImageNet-Bg (right) and the ImageNet validation set (left).

where P_n^{ood} stands for the n -th ID-relevant patches of x_i , $N_i = H \times W$ for x_i^{ood} , and $D_{train}^{all} = D_{train}^{ood} \cup D_{train}^{id}$.

OOD-global irrelevance loss. To prevent the model from misclassifying OOD samples as ID ones, we consider the global visual features for samples from D_{train}^{ood} and calculate the negative value of the entropy of the predicted probabilities across all ID categories, as the OOD-global irrelevance loss. The OOD-global irrelevance loss $\mathcal{L}_{ood}^g$ is defined as:

$$\mathcal{L}_{ood}^g = -\frac{1}{|D_{train}^{ood}|} \sum_{x \in D_{train}^{ood}} H[p(y|x)]. \quad (7)$$

Final objective. The final loss function with balanced hyperparameter λ_{ood}^g and λ_{ood}^r is as follows:

$$\mathcal{L} = \mathcal{L}_{id} + \lambda_{ood}^r \mathcal{L}_{ood}^r + \lambda_{ood}^g \mathcal{L}_{ood}^g. \quad (8)$$

OOD inference. Following [35, 36], we utilize the GL-MCM score to classify the ID and OOD samples, defined as: $S_{OOD} = \max_k p(y_k | \mathbf{x}) + \beta \max_{k,i} p(y_k | \mathbf{P}_i)$.

4. Experiments

4.1. Experimental Setup

Datasets. For ID datasets, We use ImageNet-1K [7]. For OOD datasets, we use the same as [19], including subsets of iNaturalist [47], SUN [50], Places [58], and Texture [6]. Additionally, we also use our proposed ImageNet-Bg.

The proposed background test set ImageNet-Bg. To further test the robustness of the model against background interference, we propose an ImageNet background interference test set, ImageNet-Bg, based on the ImageNet validation set with 48,285 images. All images in this dataset are generated by removing ID-relevant regions from samples in the ImageNet validation set using Eq.2. We filter the images according to Eq.3 to obtain the ImageNet-Bg(S) test set, which contains purer background information with

24,863 images. Fig. 3 shows pairs of images from the ImageNet validation set and ImageNet-Bg and the types of backgrounds where the model may take shortcuts. More examples and details are shown in *Supplementary Material*.

Evaluation metrics. For evaluation, we use the following commonly used metrics in OOD detection: Area Under the Receiver Operating Characteristic Curve (AUROC) and False Positive Rate at 95% True Positive Rate (FPR95).

Implementation details. For few-shot training, we follow the evaluation protocol established in LoCoOp [35], using 1, 2, 4, 8, and 16 shots for training and then deploy the models on the entire test sets. All pre-trained CLIP models used in our experiments are obtained from OpenCLIP [5]. For more details, please refer to the *Supplementary Material*.

Comparison methods. To demonstrate the superiority of our method, we compare it with some recent works on OOD detection based on VLMs. For training-free methods, ZOC [10], MCM [33], GL-MCM [39], CLIPN-A [49], DPM-F [57] are included. For outlier-label exposure methods, we take NegLabel [21], LAPT [55], EOE [4], OLE [8] into consideration. For methods that require few-shot training (or with fine-tuning), we consider CoOp [60], LoCoOp [35], SCT [51], DPM-T [57], ID-like [2], NegPrompt [27].

4.2. Main Results

Results on traditional OOD datasets of ImageNet-1K.

The performance on the four traditional OOD datasets of ImageNet-1K is summarized in Tab. 1. In terms of average performance, our method achieves 94.75% AUR and 25.13% FPR, representing a reduction of 1.34% in FPR and an improvement of 0.91% in AUR compared to the state-of-the-art few-shot training required method, highlighting the superiority of our method. For specific OOD datasets, our method shows significant performance improvements on SUN and Places, achieving 96.74% AUR and 18.26% FPR on SUN, and 94.01% AUR and 25.74% FPR on Places. This improvement is attributed to the larger proportion of

Method	iNaturalist		SUN		Places		Texture		Avg	
	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓
Training-free methods										
ZOC [10]	86.09	87.30	81.20	81.51	83.39	73.06	76.46	98.90	81.79	85.19
MCM [33]	94.61	30.94	92.56	37.67	89.76	44.76	86.10	57.91	90.76	42.82
GL-MCM [39]	96.71	15.18	93.09	30.42	89.90	38.85	83.63	57.93	90.83	35.47
CLIPN-A [49]	95.27	23.94	93.93	26.17	90.93	40.83	92.28	33.45	93.10	31.10
DPM-F [57] †	96.84	15.26	91.78	42.58	89.60	45.99	85.74	57.55	90.99	40.35
Outlier-label exposure methods										
NegLabel [21]	99.48	1.99	95.43	21.05	91.95	34.95	90.90	44.79	94.25	25.69
LAPT [55]	99.63	1.16	96.01	19.12	92.01	33.01	91.06	40.32	94.68	23.40
EOE [4]	97.52	12.29	95.73	20.40	92.94	30.16	85.64	57.53	92.96	30.09
OLE [8]	98.33	7.61	94.87	22.44	92.45	31.73	92.40	34.70	94.51	24.12
Requires few-shot training (or w/ fine-tuning)										
CoOp [60]	96.62	14.60	92.65	28.48	89.98	36.49	88.03	43.13	91.82	30.67
LoCoOp [35]	96.86	16.05	95.07	23.44	91.98	32.87	90.19	42.28	93.52	28.66
SCT [51]	95.86	13.94	95.33	20.55	92.24	29.86	89.06	41.51	93.37	26.47
DPM-T [57] †	97.04	14.47	93.19	33.06	89.78	39.46	87.49	49.73	91.88	34.18
ID-like [2]	98.19	8.98	91.64	42.03	91.15	41.74	94.38	26.77	93.84	29.88
NegPrompt [27] †	90.69	45.97	92.18	39.43	91.65	37.49	90.01	44.84	91.13	41.93
OSPCoOp (Ours)	97.13	15.25	96.74	18.26	94.01	25.74	91.13	41.26	94.75	25.13

Table 1. OOD detection results on ImageNet-1K as ID datasets with ViT/B-16. The boldfaced results indicate the best performance among few-shot training methods. Results marked with † are re-implemented in the 16-shot setting.

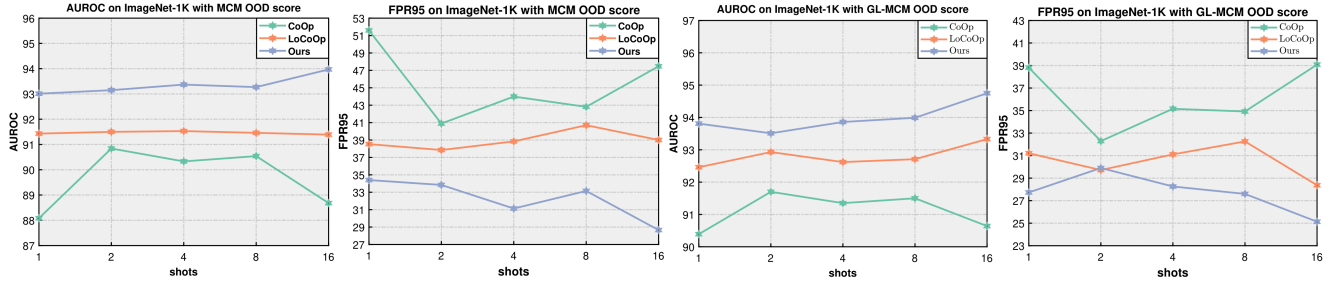


Figure 4. OOD detection results on different shot settings with MCM and GL-MCM OOD scores.

background interference in these datasets, indicating that our method effectively addresses the shortcut problem in CLIP, thereby enhancing robustness against background interference. In other OOD datasets, such as iNaturalist, our method outperforms LoCoOp, achieving a 0.27% increase in AUR and a 0.80% reduction in FPR. This further demonstrates that addressing shortcuts in the CLIP model can improve the model’s robustness across various OOD data.

Results on proposed ImageNet-Bg. The performance on the background interference OOD datasets ImageNet-Bg is summarized in Tab. 3. Notably, most methods exhibit a decline compared to using CLIP for zero-shot OOD detection (MCM, GL-MCM), and particularly CLIPN shows a significant drop, likely due to shortcuts established during pre-training on the no-text encoder. Additionally, outlier-label-exposure methods suffer from decreased OOD performance, possibly because their OOD labels inadequately

represent background interference. DPM-F improves performance by matching visual patterns. Fine-tuning methods like CoOp and LoCoOp yield only marginal improvements, indicating a struggle to mitigate the shortcut problem. Conversely, DPM-T shows a significant performance decline compared to DPM-F, suggesting that in few-shot scenarios parameter insertion to the vision encoder may worsen shortcuts and lead to overfitting on ID samples. In contrast, our method effectively addresses the shortcut issue, achieving a mean AUROC of 91.30% and a mean FPR95 of 33.84% across the two background interference OOD datasets.

Results on different shots settings. In addition to the 16-shot results shown in Tab. 1, we also test the model’s OOD detection performance under 1, 2, 4, and 8-shot settings compared with CoOp and LoCoOp, as presented in Fig. 4. In all few-shot settings, our method achieves the best performance, which fully demonstrates that overcoming shortcuts

Model	Method	iNaturalist		SUN		Places		Texture		Avg	
		AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓
ViT-B/32	LoCoOp [35]	94.67	28.08	91.33	41.45	89.17	46.60	90.35	43.71	91.20	40.05
	SCT [51]	95.24	25.01	93.19	32.64	90.55	40.16	88.44	46.67	91.86	36.12
	OSPCoOp (Ours)	96.77	16.22	95.07	25.94	92.63	31.82	90.10	39.82	93.64	28.45
RN-50	LoCoOp [35]	94.05	33.32	92.16	39.01	90.16	43.27	88.60	46.67	91.24	40.57
	SCT [51]	93.32	33.10	92.65	34.68	88.56	41.97	90.20	37.13	91.18	36.72
	OSPCoOp (Ours)	95.04	24.96	95.12	24.96	91.52	34.80	90.15	45.78	93.11	32.12

Table 2. Results on ImageNet-1K as ID datasets with different CLIP architecture in 16-shot setting.

Method	ImageNet-Bg(S)		ImageNet-Bg	
	AUROC↑	FPR95↓	AUROC↑	FPR95↓
Training-free methods				
MCM [33]	88.64	44.93	88.85	44.25
GL-MCM [39]	87.71	45.96	87.75	44.59
CLIPN-A [49]	81.61	66.87	81.67	66.05
DPM-F [57]	89.36	41.88	89.53	40.78
NegLabel [21]	85.41	59.56	85.26	59.53
LAPT [55]	84.17	56.78	84.34	55.33
OLE [8]	84.48	62.27	84.70	61.31
Requires few-shot training (or w/ fine-tuning)				
CoOp [60]	89.06	41.36	89.41	42.23
LoCoOp [35]	89.88	38.34	90.01	37.70
DPM-T [57]	85.35	52.15	85.46	51.86
NegPrompt [27]	88.11	49.51	88.09	49.49
SCT [51]	89.30	39.79	89.39	39.25
OSPCoOp (Ours)	91.22	34.09	91.37	33.58

Table 3. Results on ImageNet-Bg as OOD and ImageNet-1K as ID. The boldfaced results indicate the best performance.

is beneficial for OOD detection with VLMs, especially in few-shot scenarios. Notably, even in the 1-shot setting, our method still performs well, achieving a mean AUROC of 93.81% and a mean FPR95 of 27.73%.

4.3. Ablation Study

Backbones. We test the performance of our methods with different CLIP backbones, as shown in Tab. 2. Results show that our method can achieve good performance even with the CNN-based backbones. Overall, the more advanced the backbone used, the better our method performs.

OOD augmentation. To verify the effectiveness of decoupling ID/OOD content and performing OOD data augmentation, we conduct experiments with different OOD augmentation strategies, as shown in Tab. 4. 'None' indicates that no OOD data is used, meaning that only the ID data's OOD regions participate in the optimization of the loss λ_{ood}^r . 'Bg' and 'Rep' respectively denote the OOD data constructed using Eq. 2 and the OOD data generated by repeating local ID regions. Using 'Bg' and 'Rep' data improves AUR by 1.68% and 0.46% respectively. This in-

OOD Aug	Places		Texture		Avg	
	AUR↑	FPR↓	AUR↑	FPR↓	AUR↑	FPR↓
None	92.02	34.71	89.27	46.38	92.97	32.54
Rep	92.58	32.17	90.08	44.50	93.43	29.82
Bg	93.85	25.95	90.90	41.35	94.65	24.77
Rep+Bg	94.01	25.74	91.13	41.26	94.75	25.13

Table 4. Ablation study of OOD augmentation. 'Rep' stands for the data generated by repeating local ID regions, and 'Bg' stands for using the decoupled OOD content with ID regions inpainted. 'Avg' stands for the results on the four traditional OOD datasets.

icates that using background as pseudo-OOD supervision can enhance the model's robustness to OOD samples. We also test the impact of ϵ in the filter, as shown in Fig. 6a. A larger ϵ is more stringent and can yield purer background OOD samples, while a smaller ϵ can provide more OOD samples. Results show that balancing the quantity and quality of OOD augmentation can lead to better performance. Fig. 5 visualizes the OOD Aug samples and ID samples. The OOD samples partly show a clustering trend, with a small portion distributed at the edge of the ID samples.

ID augmentation. Under different shot settings, we conduct experiments on ID augmentation. The average results on the four traditional OOD datasets are shown in Tab. 5, demonstrating that using ID augmentation with fewer shots results in a remarkable performance boost. The AUROC is improved by 0.11%, 0.40%, and 0.30% on one-shot, two-shot, and four-shot settings respectively. For more details,

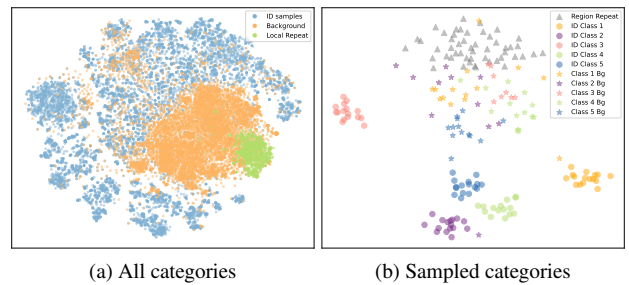


Figure 5. Visualization of ID samples and OOD samples.

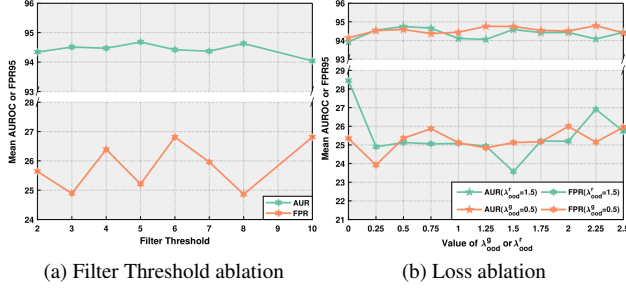


Figure 6. Ablation study of OOD augmentation and loss.

please refer to *Supplementary Material*.

ID Aug	One-shot		Two-shot		Four-shot	
	AUR ↑	FPR ↓	AUR ↑	FPR ↓	AUR ↑	FPR ↓
×	93.81	27.73	93.51	29.90	93.86	28.26
✓	93.92	27.77	93.91	27.71	94.16	26.40

Table 5. Ablation study of ID augmentation with GL-MCM score.

Loss function. The results of the ablation experiments on λ_{ood}^r and λ_{ood}^g in the loss function are shown in Tab 6. It can be found with ease that when only using $\mathcal{L}_{ood}^r$, AUR is increased by 1.89% and when only using $\mathcal{L}_{ood}^g$, AUR is increased by 2.09%. By optimizing both of the losses, AUR achieved a significant improvement of 2.70%, indicating that using constraints on regions and using pseudo-OOD supervision both contribute positively. Fig. 6b shows the performance variation with respect to λ_{ood}^r and λ_{ood}^g , which shows that our method is insensitive to them.

OOD scores. Fig. 4 shows the performance with MCM and GL-MCM OOD scores. Results indicate that our method achieves better performance with different OOD scores.

4.4. Analysis

Comparison with LoCoOp. Similar to ours, LoCoOp extracts ID-irrelevant regions and optimizes the region logits of CLIP. However, it relies on CLIP feature similarity to delineate these regions, which may misclassify background areas as ID-relevant due to the shortcut problem. In contrast, our method uses masks for constraints, effectively refining the CLIP model’s attention mechanism. Tab. 7 compares the ID region delineation methods of LoCoOp and our method. Our region delineation method improved OOD

λ_{ood}^r	λ_{ood}^g	AUROC↑	FPR95↓
×	×	92.05	32.78
✓	×	93.94	28.45
×	✓	94.14	25.35
✓	✓	94.75	25.13

Table 6. Ablation study of loss on ImageNet with ViT-B-16.

Methods	Places		Texture		Avg	
	AUR ↑	FPR ↓	AUR ↑	FPR ↓	AUR ↑	FPR ↓
Rank	93.56	28.23	90.52	42.98	94.44	26.41
Mask+Rank	93.70	28.24	90.97	39.80	94.52	25.90
Mask	94.01	25.74	91.13	41.26	94.75	25.13

Table 7. Comparison of different region delineation methods. ‘RANK’ refers to the method of LoCoOp, ‘Mask’ denotes ours, and ‘RANK+Mask’ indicates using feature similarity of CLIP for those images without mask. ‘Avg’ stands for the results on the four traditional OOD datasets.

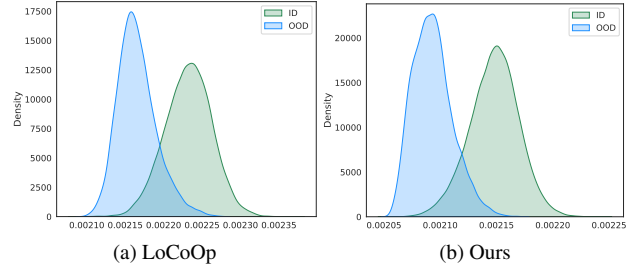


Figure 7. ID/OOD score visualization: LoCoOp vs. Ours on SUN

performance by 0.31% while introducing ‘RANK’ into our method resulted in a decline of 0.23%. This suggests that using CLIP feature similarity for region delineation can mislead the model’s focus toward background areas, which strengthens shortcuts and adversely affects OOD performance. Fig. 7 visualizes the ID/OOD score distribution of our method compared to LoCoOp, demonstrating that our method establishes a better ID/OOD decision boundary.

5. Conclusions and Limitations

In this paper, we propose OSPCoOp for OOD detection in VLMs, specifically focusing on the shortcut problem. We find that CLIP often relies on background information, which can lead to failures in OOD detection, especially for background interference. OSPCoOp effectively separates ID-irrelevant regions from ID-relevant ones, allowing the model to minimize the local logits associated with ID-irrelevant regions and achieve robust OOD detection. Additionally, we present ImageNet-Bg, a novel OOD evaluation benchmark designed to facilitate a comprehensive assessment of model robustness against background interference. Our results show that decoupling background information to generate pseudo-OOD supervision, constraining the model’s responses to ID-irrelevant regions, can effectively mitigate the shortcut problem. We hope our work will inspire further research into OOD detection in VLMs.

Limitation. The proposed ImageNet-bg remains limited in scale, and theoretical analysis is also needed in future work.

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